

Motion in the Schwarzschild geometry

22

*We are merely the stars' tennis-balls, struck and banded
Which way to please them*
John Webster (1580–1625) *The Duchess of Malfi*

Thanne longen folk to goon on pilgrimages
Geoffrey Chaucer (c.1343–1400)
Prologue to the Canterbury Tales

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We found in the last chapter that the Schwarzschild line element is given by

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (22.2)$$

This equation represents the metric field outside a spherically symmetric object, such as a star, of mass M . In this chapter and the following ones, we shall examine the free-falling motion of particles in this metric field, where the particles follow the geodesics of this geometry. We will find that general relativity allows a richer range of possible motions than are found in the Newtonian problem, including, in addition to the interesting trajectories of massive particles, effects on the motion of photons: the particles of light.

At first glance the problem of relativistic motion looks like a formidable one involving deriving solutions to the geodesic equation,¹ with terms provided by the connection coefficients found from eqn 22.2. However, just as problems involving Newtonian orbits are greatly simplified by knowing about conserved quantities, we shall see how an analysis of conserved quantities, along with simple facts about the velocity $\mathbf{u}$, allow us to solve a variety of key problems. In particular, we make a lot of use of the identity² $\mathbf{u} \cdot \mathbf{u} = -1$ for massive particles and $\mathbf{u} \cdot \mathbf{u} = 0$ for massless ones. In addition, just as energy and angular momentum are conserved in the Newtonian potential, we will see that something very much like these is also conserved in the relativistic case, as we might naively expect. We turn first, therefore, to the method for extracting conserved quantities.

In this chapter, we'll need to use the components of the Schwarzschild metric

$$\begin{aligned} g_{tt} &= - \left(1 - \frac{2M}{r}\right), \\ g_{rr} &= \left(1 - \frac{2M}{r}\right)^{-1}, \\ g_{\theta\theta} &= r^2, \\ g_{\phi\phi} &= r^2 \sin^2 \theta. \end{aligned} \quad (22.1)$$

¹This is the differential equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0,$$

with the connection coefficients determined from the metric.

²Reminder: A massive particle's world line is parametrized by proper time so that, in a coordinate system with $x^\mu = (t, r, \theta, \phi)$, the velocity $\mathbf{u}$ has components $u^\mu = (u^t, u^r, u^\theta, u^\phi)$, i.e.

$$u^\mu = \left(\frac{dt}{d\tau}, \frac{dr}{d\tau}, \frac{d\theta}{d\tau}, \frac{d\phi}{d\tau} \right).$$

↪ Killing vectors are discussed in detail in Chapter 33 in the context of geometry.

22.1 Constants of the motion

One of the most important factors in analysing motion in mechanics is the identification of constants of the motion. In geometrical problems, exemplified by the physics of gravitation, conserved quantities can be identified by finding a set of fields known as the **Killing vector fields** ξ of the metric. These are very simply identified by inspection, by noting which coordinates do not feature in the components of the metric. For now we will give (but not prove) a simple recipe for identifying conserved quantities.

To find a conserved quantity from a metric field:

Step 1: Look at the metric components and identify those coordinates on which none of the metric components depend.

Step 2: If the variable x^α does not feature, we have a **Killing vector** $\xi = e_\alpha$.

Step 3: Particles travelling along a geodesic have a velocity $\mathbf{u} = u^\mu \mathbf{e}_\mu$, which is the tangent to a geodesic. The quantity

$$\xi \cdot \mathbf{u} = g_{\mu\nu} \xi^\mu u^\nu \quad (22.3)$$

is **conserved** along the geodesic.

To identify the Killing vectors for the Schwarzschild geometry, we spot (**step 1**) that the metric components are all independent of the coordinates t and ϕ . Ordering the coordinates (t, r, θ, ϕ) we identify two corresponding Killing vectors (**step 2**). First, from the independence of t we extract a Killing vector ξ via

$$\xi = e_t, \quad \xi^\mu = (1, 0, 0, 0), \quad (\text{time independence}), \quad (22.4)$$

and from the independence of ϕ we identify a Killing vector η as we have

$$\eta = e_\phi, \quad \eta^\mu = (0, 0, 0, 1), \quad (\text{rotational symmetry}). \quad (22.5)$$

We now use the fact that $\xi \cdot \mathbf{u}$ is a constant along a geodesic (**step 3**) to identify the conserved quantities. The first is identified as³

$$\xi \cdot \mathbf{u} = u^\mu \mathbf{e}_\mu \cdot \mathbf{e}_t = u^\mu g_{\mu t} = u_t, \quad (22.6)$$

or equivalently

$$\xi \cdot \mathbf{u} = g_{tt} u^t = - \left(1 - \frac{2M}{r} \right) u^t = - \left(1 - \frac{2M}{r} \right) \frac{dt}{d\tau}, \quad (22.7)$$

which is conventionally named $-\tilde{E}$, which is to say we have a conserved quantity

$$\tilde{E} = -\xi \cdot \mathbf{u} = \left(1 - \frac{2M}{r} \right) u^t. \quad (22.8)$$

This quantity $\tilde{E}$ tells us about the energy per unit rest mass.⁴ In general, it can be interpreted, for timelike geodesics, as total energy per unit rest mass, measured by a static observer at infinity.

³Remember the very useful fact that $g_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu$.

⁴In general, we can compute the energy of a particle with momentum $\mathbf{p}$, measured by an observer with velocity $\mathbf{u}$ by computing $E = -\mathbf{p} \cdot \mathbf{u}$. Notice that an observer at rest at infinity in the field of a spherical star has, in the rest frame of the star (which is also the observer's rest frame), a velocity $\mathbf{u}_{\text{obs}}$ with components $u^\mu = (1, 0, 0, 0)$, so that $\mathbf{u}_{\text{obs}} = \xi$. (This follows because spacetime is flat at infinity. In general, u^t for an observer is determined in their rest frame using the condition $\mathbf{u} \cdot \mathbf{u} = -1$.) Consider a particle with momentum $\mathbf{p}$ travelling along a geodesic in the field of a spherical mass distribution. Its energy, measured locally at infinity by an observer at rest, is therefore $E_\infty = -\mathbf{p} \cdot \xi$. We will use this result below.

The other conserved quantity, called $\tilde{L}$ is also straightforward to pick out as $\boldsymbol{\eta} \cdot \mathbf{u} = u_\phi$, or, from $u_\phi = g_{\phi\phi} u^\phi$

$$\tilde{L} = \boldsymbol{\eta} \cdot \mathbf{u} = r^2 \sin^2 \theta u^\phi. \quad (22.9)$$

We interpret this as angular momentum per unit rest mass. Note that for equatorial motion (i.e. with $\theta = \pi/2$) this becomes $\tilde{L} = r^2 \dot{\phi}$, which looks just like the expression for angular momentum per unit mass in cylindrical coordinates.

So far we have considered only the geometry, and not what sort of particle is in free fall. This information is supplied via the square of the velocity vector $\mathbf{u} \cdot \mathbf{u}$. Massive particles move along timelike geodesics and so we fix the magnitude of the velocity vector (which is tangent to the geodesic) by saying

$$\mathbf{u} \cdot \mathbf{u} = g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -1, \quad (\text{massive particles}), \quad (22.10)$$

where the proper time τ has been employed as an affine parameter marking off the trajectory of the particle. Photons travel along null geodesics, whose velocity tangent vector has the property

$$\mathbf{u} \cdot \mathbf{u} = g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0, \quad (\text{massless photons}), \quad (22.11)$$

where λ is an affine parameter for the photon world line.

22.2 Gravitational redshift

With these preliminaries, we can derive some results. The first is a result for photons: the gravitational redshift caused by the Schwarzschild geometry.⁵ To analyse this, we consider the motion of a photon with null momentum $\mathbf{p}$ propagating along a radial line. By symmetry, this is a geodesic. The photon has the property⁶ that $\mathbf{p} \cdot \boldsymbol{\xi}$ is conserved along the geodesic, which implies that, in this geometry, $g_{tt}p^t = (1 - 2M/r)p^t(r)$ is conserved as r is varied (rather than the flat-space energy p^t , as we might naively have expected). Remember, however, that an observer at r does not measure p^t ; they measure the local value $p^{\hat{t}} = \hbar\omega$.

We can find out how the photon's measured frequency ω changes as the photon moves radially outwards from a massive star. We consider two static observers, one at radius $r = R$ and one at $r = \infty$ (Fig. 22.1). Being observers, their world lines have tangents that are timelike velocity vectors. The energy of a photon with momentum $\mathbf{p}$, measured by an observer⁷ with velocity $\mathbf{u}_{\text{obs}}$ is

$$\hbar\omega = -\mathbf{p} \cdot \mathbf{u}_{\text{obs}}. \quad (22.12)$$

If the observer is at rest with respect to the star, then $u_{\text{obs}}^i = 0$ and we can determine the timelike component of $\mathbf{u}_{\text{obs}}$ using the velocity identity

$$g_{tt} (u_{\text{obs}}^t)^2 = -1. \quad (22.13)$$

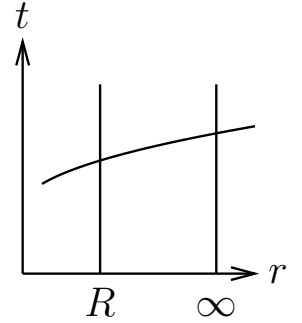


Fig. 22.1 A photon world line meets an observer at R and one at infinity.

⁵Here we revisit our second cause of frequency shifts discussed earlier in Chapter 6. (The others are the Doppler effect and the cosmological redshift.)

⁶This follows since we shall assume $\mathbf{p}$ to be parallel to $\mathbf{u}$ for light and $\mathbf{p}$ is therefore tangent to the geodesics. (For a more careful discussion of photon momentum, see Section 24.7.)

⁷An alternative rule would be to use the vielbein components to say that the measured energy $p^{\hat{t}}$ is related to p^t via

$$p^{\hat{t}}(r) = (e_\mu)^{\hat{t}}(r)p^\mu(r).$$

This alternative approach is examined in the exercises. Here, instead, since we are trying to relate measurements made by local observers in two places, we use the conservation laws encoded in the Killing vector $\boldsymbol{\xi}$ (i.e. that $\mathbf{u} \cdot \boldsymbol{\xi}$ is conserved along a geodesic). Our strategy is therefore to incorporate the energy Killing vector $\boldsymbol{\xi}$ into the description of the stationary observer's velocity.

With the Schwarzschild metric component g_{tt} , this gives a velocity component at a radius r of

$$u_{\text{obs}}^t(r) = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}, \quad (22.14)$$

⁸Recall that ξ has components $\xi^\mu = (1, 0, 0, 0)$.

or in terms of one of the Killing vectors,⁸ $\mathbf{u}_{\text{obs}}(r) = (1 - 2M/r)^{-\frac{1}{2}}\xi$, which is the key equation. This expression means we can write eqn 22.12 in terms of the Killing vector ξ . Specifically, the energy measured by the stationary observer sat at a radius R is

$$\hbar\omega_R = \left(1 - \frac{2M}{R}\right)^{-\frac{1}{2}} (-\xi \cdot \mathbf{p})_R, \quad (22.15)$$

where subscript R implies the quantity is evaluated at this radius. For the energy measured by the observer at infinity, the factor $(1 - 2M/r)^{-\frac{1}{2}} \rightarrow 1$ and so $\hbar\omega_\infty = (-\xi \cdot \mathbf{p})_\infty$. However, since the photon is moving along a geodesic, the quantity $\xi \cdot \mathbf{p}$ is conserved in the motion, and so $(-\xi \cdot \mathbf{p})_R = (-\xi \cdot \mathbf{p})_\infty$ and we must therefore have

$$\omega_\infty = \omega_R \left(1 - \frac{2M}{R}\right)^{\frac{1}{2}}. \quad (22.16)$$

The frequency at infinity is less than the frequency at the point R (see Fig. 22.2). One can rationalize this result by saying that the photon's energy is lowered through its climbing of the gravitational potential.

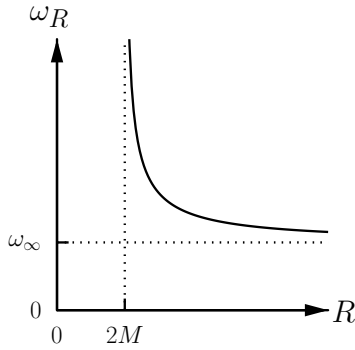


Fig. 22.2 A plot of ω_R against R for $R > 2M$.

We can summarize the important results that we shall use to compute trajectories:

■ For massive particles, the particle velocity $\mathbf{u}$ is tangent to the world line and has the property $\mathbf{u} \cdot \mathbf{u} = -1$.

■ For photons we have $\mathbf{u} \cdot \mathbf{u} = 0$. The interval $ds = 0$ on the world line provides a useful constraint.

■ We compute conservation laws using the Killing rule, that says that if the metric components are independent of variable x^α , the component u_α is conserved along a geodesic.

■ To access the coordinate velocity dx^i/dt we use the trick

$$\frac{dx^i}{dt} = \frac{dx^i}{d\tau} \frac{d\tau}{dt} = \frac{1}{u^0} \frac{dx^i}{d\tau}. \quad (22.17)$$

⁹The approach described here is covered in many books, but beware that conventions for defining some of the terms differ. We follow the approach and notation used in Hartle.

22.3 Motion in Schwarzschild spacetime

We now turn to the motion of massive particles in the Schwarzschild geometry.⁹ It might be assumed that we need to solve the full geodesic equation to understand the motion of particles and photons. Fortunately for us, the only ingredients needed are the constants of the motion given by the Killing vectors, and the velocity identity $\mathbf{u} \cdot \mathbf{u} = -1$.

Example 22.1

For simplicity, we assume motion in the equatorial plane of the geometry, which is to say that we fix $\theta = \pi/2$ and so $u^\theta = 0$. To analyse motion of a particle we start by using the velocity identity for the particle $\mathbf{u} \cdot \mathbf{u} = g_{\mu\nu}u^\mu u^\nu = -1$, to write

$$-\left(1 - \frac{2M}{r}\right)(u^t)^2 + \left(1 - \frac{2M}{r}\right)^{-1}(u^r)^2 + r^2(u^\phi)^2 = -1. \quad (22.18)$$

This can be rewritten using the conserved quantities $\tilde{E}$ and $\tilde{L}$ as

$$-\left(1 - \frac{2M}{r}\right)^{-1}\tilde{E}^2 + \left(1 - \frac{2M}{r}\right)^{-1}\left(\frac{dr}{d\tau}\right)^2 + \frac{\tilde{L}^2}{r^2} = -1. \quad (22.19)$$

A little massaging of the above expression gives

$$\frac{\tilde{E}^2 - 1}{2} = \frac{1}{2}\left(\frac{dr}{d\tau}\right)^2 + \frac{1}{2}\left[\left(1 - \frac{2M}{r}\right)\left(1 + \frac{\tilde{L}^2}{r^2}\right) - 1\right]. \quad (22.20)$$

Now we introduce a new constant energy-like variable,¹⁰ $\mathcal{E} = (\tilde{E}^2 - 1)/2$, and define an effective potential for motion in the Schwarzschild geometry of

$$V_{\text{eff}}(r) = \frac{1}{2} \left[\left(1 - \frac{2M}{r}\right) \left(1 + \frac{\tilde{L}^2}{r^2}\right) - 1 \right] = -\frac{M}{r} + \frac{\tilde{L}^2}{2r^2} - \frac{M\tilde{L}^2}{r^3}, \quad (22.21)$$

and we end up with the familiar equation $\mathcal{E} = \frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 + V_{\text{eff}}(r)$.

The result of the last example is that the motion of a massive particle obeys the effective-potential equation

$$\mathcal{E} = \frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 + V_{\text{eff}}(r), \quad (22.22)$$

with $\mathcal{E} = (\tilde{E}^2 - 1)/2$ and

$$V_{\text{eff}}(r) = -\frac{M}{r} + \frac{\tilde{L}^2}{2r^2} - \frac{M\tilde{L}^2}{r^3}. \quad (22.23)$$

This is similar to the equation for motion in an effective potential that we saw¹¹ in Chapter 20 for Newtonian motion. We conclude that motion takes place in an effective potential V_{eff} . The potential for the Schwarzschild geometry is shown in Fig. 22.3. Compared to the Newtonian potential, this one has more structure: specifically, with increasing r , an initial increase in V_{eff} to a maximum, followed by behaviour that looks similar to its Newtonian cousin. This new structure leads to the richness of the new trajectories allowed by relativity.

Example 22.2

We can restore factors of c and G to find

$$V_{\text{eff}} = \frac{1}{c^2} \left(-\frac{GM}{r} + \frac{\tilde{L}^2}{2r^2} - \frac{GM\tilde{L}^2}{c^2 r^3} \right). \quad (22.24)$$

Now define the non-relativistic, Newtonian energy E_N via

$$\tilde{E} = \frac{mc^2 + E_N}{mc^2}. \quad (22.25)$$

This allows us to write

$$E_N = \frac{m}{2} \left(\frac{dr}{d\tau}\right)^2 + \frac{L^2}{2mr^2} - \frac{GMm}{r} - \frac{GML^2}{mc^2 r^3}, \quad (22.26)$$

where $L = m\tilde{L}$. We conclude that the usual Newtonian energy equation is augmented at order $1/c^2$ by the relativistic, $1/r^3$ term.

We are now in the position to be able to provide a description of the motion. If we are interested in what would be measured by an observer, we will need access to the vielbein components for the Schwarzschild geometry in order to shift into the orthonormal frame of the observer. The vielbein for a stationary observer is given in the margin. It is worth noting that for this observer at infinity, measurements of the $\hat{t}$ and $\hat{r}$ components of a vector will give the coordinate values as both of the relevant vielbein components are unity at infinity.

¹⁰Particles coming in from infinity have $\tilde{E} > 1$ and so have effective energy $\mathcal{E} > 0$.

¹¹The equations of motion differ by a factor of m . The Newtonian version has an effective potential energy U_{eff} and so an effective potential

$$V_{\text{eff}}(r) = \frac{U_{\text{eff}}(r)}{m} = -\frac{M}{r} + \frac{\tilde{L}^2}{2r^2}.$$

The relativistic version has an extra term with a $1/r^3$ dependence.

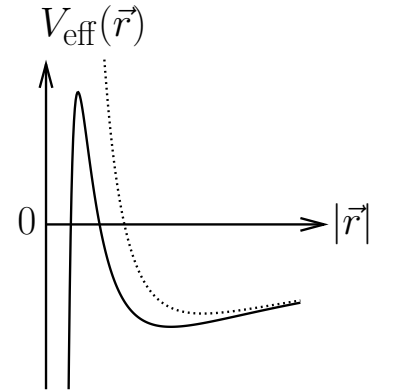


Fig. 22.3 The relativistic effective potential for a particle (solid line, with $\tilde{L}/M = 4.3$), compared to the Newtonian one (dotted line).

The vielbein components for an observer in the orthonormal frame (where the observer is stationary with respect to the coordinate frame) has components

$$\begin{aligned} (e_t)^{\hat{t}} &= \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}}, \\ (e_r)^{\hat{r}} &= \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}, \\ (e_\theta)^{\hat{\theta}} &= r, \\ (e_\phi)^{\hat{\phi}} &= r \sin \theta. \end{aligned}$$

It is important to remember that if we want to shift into the frame of a moving observer we will need a different vielbein. An example we have seen before that we will use several times is the freely falling observer's frame.

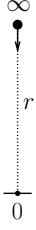


Fig. 22.4 The radial plunge.

¹²At infinity, spacetime is flat and the particle is at rest. Initially, the components of the velocity must therefore be $u^\mu(r = \infty) = \frac{dx^\mu}{d\tau} = (1, 0, 0, 0)$. Recall also that the energy

$$\tilde{E} = \left(1 - \frac{2M}{r}\right) u^t$$

is constant along a geodesic. For $r \rightarrow \infty$ we have $\tilde{E} = 1$. Also recall that the (constant) effective energy is

$$\mathcal{E} = \frac{\tilde{E}^2 - 1}{2} = 0.$$

¹³There is no mechanism to pick up angular velocity, hence the zeros for the θ and ϕ components of velocity.

¹⁴The falling observer's local frame is flat, but moving with respect to the coordinate frame, so differs from the conventional orthonormal frame we often use. To find this frame, note that we always take $e_{\hat{t}} = \mathbf{u}$. Then, for convenience choose $(e_{\hat{\theta}})^\mu = (0, 0, 1, 0)$ and $(e_{\hat{\phi}})^\mu = (0, 0, 0, 1)$. This means that the one remaining vielbein component is

$$(e_{\hat{r}})^\mu = \begin{pmatrix} \frac{-\sqrt{2M/r}}{1-2M/r} \\ 1 \\ 0 \\ 0 \end{pmatrix}. \quad (22.30)$$

¹⁵In this section, it is necessary to fix the value of r at some value of the proper time τ in order to fix the integration constant.

¹⁶A useful quantity to note for computations is the coordinate velocity

$$\frac{dr}{dt} = -\left(1 - \frac{2M}{r}\right) \left(\frac{2M}{r}\right)^{\frac{1}{2}}. \quad (22.34)$$

The magnitude of this quantity is the coordinate escape velocity of a particle, as discussed in the exercises.

22.4 Example: the radial plunge

Let's consider a massive particle that starts at rest at infinity and plunges towards a star along a radial line (Fig. 22.4). This line is a geodesic.

Example 22.3

We have immediately that the angular momentum $\tilde{L} = 0$ since the plunge is radial. 'At rest at infinity' implies that at a great distance from the gravitating object we have¹² $u^t = dt/d\tau = 1$ and so, for the duration of the freefall, $\tilde{E} = 1$ and $\mathcal{E} = 0$. The fact that $\tilde{E} = 1$ allows us to pick out, at later times, that

$$u^t = \frac{dt}{d\tau} = \left(1 - \frac{2M}{r}\right)^{-1}. \quad (22.27)$$

The effective energy $\mathcal{E}$ in eqn 22.22 for a radial plunge with $\tilde{L} = 0$ and $\mathcal{E} = 0$ gives us an equation of motion

$$0 = \frac{1}{2} \left(\frac{dr}{d\tau}\right)^2 - \frac{M}{r}, \quad (22.28)$$

which provides us with an expression for the radial velocity $u^r = dr/d\tau = -(2M/r)^{\frac{1}{2}}$, where the sign is chosen so that the particle is falling towards the gravitating star. We conclude that the velocity during the fall is given by a vector $\mathbf{u}$ with components in the coordinate frame of¹³

$$u^\mu = \left(\left(1 - \frac{2M}{r}\right)^{-1}, -\left(\frac{2M}{r}\right)^{\frac{1}{2}}, 0, 0 \right). \quad (22.29)$$

The vielbein components for this observer's frame can also be computed.¹⁴

Let's consider how much proper time and coordinate time elapses for the falling particle.

Example 22.4

First the proper time. We rewrite eqn 22.28 as an integral

$$\int_r^0 r^{\frac{1}{2}} dr = - \int_{\tau}^{\tau_*} (2M)^{\frac{1}{2}} d\tau, \quad (22.31)$$

where the negative square root has been chosen for an inward-travelling particle and τ_* fixes the value of the proper time¹⁵ at the end of the plunge, when $r = 0$. This can be integrated with the result that

$$r(\tau) = \left(\frac{3}{2}\right)^{\frac{2}{3}} (2M)^{\frac{1}{3}} (\tau_* - \tau)^{\frac{2}{3}}. \quad (22.32)$$

We can then say that the proper time for a falling astronaut is given by

$$\tau(r) = \tau_* - \left(\frac{2}{3}\right) \left(\frac{1}{2M}\right)^{\frac{1}{2}} r^{\frac{3}{2}}. \quad (22.33)$$

The variation of τ with the Schwarzschild radius coordinate r is shown in Fig. 22.5. Notice that, from an arbitrary starting value of r it takes a finite amount of proper time to reach any value of r , as we might expect.

Next, the amount of elapsed coordinate time t . Since $(e_t)^{\hat{t}} = 1$ at infinity, this is also the time measured by an observer at infinity watching the particle plunge. We have¹⁶

$$\frac{dt}{dr} = \frac{dt}{d\tau} \frac{d\tau}{dr} = \frac{u^t}{u^r} = -\left(\frac{2M}{r}\right)^{-\frac{1}{2}} \left(1 - \frac{2M}{r}\right)^{-1}. \quad (22.35)$$

This equation can be integrated to give

$$t = t_* + 2M \left[-\frac{2}{3} \left(\frac{r}{2M} \right)^{\frac{3}{2}} - 2 \left(\frac{r}{2M} \right)^{\frac{1}{2}} + \ln \left| \frac{\left(\frac{r}{2M} \right)^{\frac{1}{2}} + 1}{\left(\frac{r}{2M} \right)^{\frac{1}{2}} - 1} \right| \right], \quad (22.36)$$

where t_* gives the time coordinate when $r = 0$. This function is graphed in Fig. 22.6.

There are some curious facts about the results from the last example: the most notable being that, from the point of view of the observer at infinity, it takes an *infinite* amount of coordinate time for an in-falling astronaut to reach $r = 2M$ (but a finite proper time). We will return to this point in Chapter 25.

Finally, let's examine the coordinate velocity of the particle that plunges radially in a Schwarzschild metric.

Example 22.5

We won't assume the particle starts from rest this time. The velocity identity gives us

$$\begin{aligned} -1 &= \mathbf{u} \cdot \mathbf{u} \\ &= - \left(1 - \frac{2M}{r} \right) (u^t)^2 + \left(1 - \frac{2M}{r} \right)^{-1} (u^r)^2 \\ &= \left[- \left(1 - \frac{2M}{r} \right) + \left(1 - \frac{2M}{r} \right)^{-1} \left(\frac{dr}{dt} \right)^2 \right] (u^t)^2 \\ &= \left[- \left(1 - \frac{2M}{r} \right) + \left(1 - \frac{2M}{r} \right)^{-1} \left(\frac{dr}{dt} \right)^2 \right] (u^t)^2 \left(1 - \frac{2M}{r} \right)^{-2}, \end{aligned} \quad (22.37)$$

where, in the final line we've written the equation in terms of the constant of the motion $\mathbf{u} \cdot \boldsymbol{\xi} = u_t$. Solving for $(dr/dt)^2$ we find

$$\left(\frac{dr}{dt} \right)^2 = \left(1 - \frac{2M}{r} \right)^2 \left[1 - \left(1 - \frac{2M}{r} \right) \frac{1}{(u_t)^2} \right]. \quad (22.38)$$

This coordinate velocity vanishes as the particle approaches $r = 2M$ (Fig. 22.7).

What does a stationary observer determine as the coordinate velocity? We won't assume that they are at infinity here and so we shift to the orthonormal frame using the vielbein. Remember that for 1-forms like $d\hat{t}$ and $d\hat{r}$ we have $\omega^{\hat{\mu}} = (e_{\mu})^{\hat{\mu}} \omega^{\mu}$. We find

$$\begin{aligned} d\hat{t} &= \left(1 - \frac{2M}{r} \right)^{\frac{1}{2}} dt, \\ d\hat{r} &= \left(1 - \frac{2M}{r} \right)^{-\frac{1}{2}} dr. \end{aligned} \quad (22.39)$$

Noting that the derivative $dr/dt \equiv d\mathbf{r}/d\mathbf{t}$, we have that the stationary observer observes a plunge with a velocity

$$\frac{d\hat{r}}{d\hat{t}} = \left(1 - \frac{2M}{r} \right)^{-1} \frac{dr}{dt}. \quad (22.40)$$

Irrespective of u_t , and hence the initial velocity, this locally measured velocity approaches unity or, in real units the speed of light, as $r \rightarrow 2M$ (Fig. 22.7).

Using only the velocity identity and the conservation of energy and angular momentum, we have shown a number of curious relativistic effects in a particle's motion. We'll pick up these points in a few chapters' time. In the next chapter, we turn to orbits.

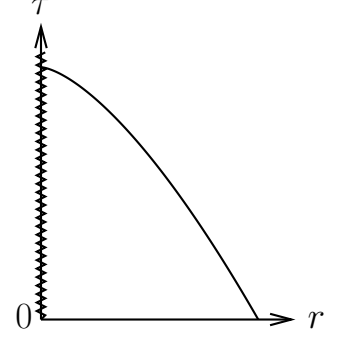


Fig. 22.5 The evolution of the coordinate r with the proper time τ for the radial plunge.

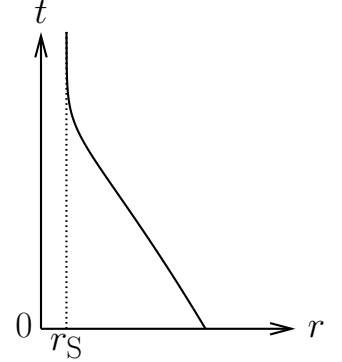


Fig. 22.6 The evolution of the coordinate r with the coordinate time t for the radial plunge. The dotted line shows $r_S = 2M$.

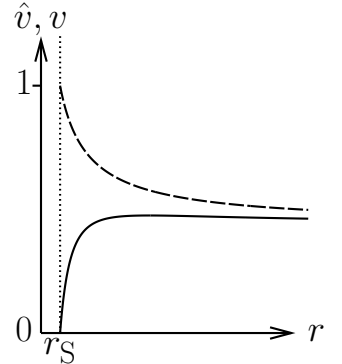


Fig. 22.7 Example evolution of the coordinate velocity $v = dr/dt$ with radial coordinate r for the radial plunge (solid line). The dashed line shows the coordinate velocity $\hat{v} = d\hat{r}/d\hat{t}$ measured by a stationary observer.

Chapter summary

- Motion in the Schwarzschild geometry can often be computed using the velocity $\mathbf{u}$, subject to the constraint that $\mathbf{u} \cdot \mathbf{u} = -1$. The constants of the motion $\tilde{E}$ and $\tilde{L}$ follow from the independence of the metric components from coordinates t and ϕ .
- The effective potential method allows motion to be analysed in terms of a relativistic V_{eff} .
- A particle that plunges from rest at infinity has $\mathcal{E} = 0$ and a velocity $\mathbf{u}$ with components

$$u^\mu = \left(\left(1 - \frac{2M}{r}\right)^{-1}, -\left(\frac{2M}{r}\right)^{\frac{1}{2}}, 0, 0 \right). \quad (22.41)$$

Exercises

- (22.1) Consider the acceleration $\mathbf{a} = \nabla_{\mathbf{u}}\mathbf{u}$. By the geodesic equation, the components of the acceleration $a^\mu = u^\alpha u^\mu{}_{;\alpha}$ vanish in free fall.
- (a) What are the components of the acceleration for a particle at rest in a static gravitational field with metric components $g_{\mu\nu}$?
- (b) What are the components of the acceleration of a particle at rest at a fixed value of r , in a Schwarzschild coordinate system.
- (c) What is the proper acceleration α (i.e. the acceleration measured in the frame of the particle)?
- (22.2) *Although we have used a variety of shortcuts to compute the equations of motion, the same information is available from the geodesic equation.*
- (a) Compute the components of the geodesic equation for the Schwarzschild geometry with line element given in the form of eqn 22.2.
- If you've done this previously for the line element expressed in different variables, you could simply re-express the equations in terms of (t, r, θ, ϕ) .*
- (b) How do these expressions simplify for motion in the plane with $\theta = \pi/2$?
- (c) Show that the equation of motion for the r coordinate is consistent with eqn 22.22.
- (22.3) Derive eqn 22.16 using the vielbein components, and the fact that p_t is conserved on the geodesic.
- (22.4) Verify eqn 22.26 in the Newtonian limit.
- (22.5) A stationary observer on the surface of a spherical mass M of radius R launches a projectile at its escape velocity. It does not experience any force during its motion and so it follows a geodesic.
- (a) What is the escape velocity of the projectile measured in the observer's rest frame?
- (b) What is the energy of the projectile as measured by the observer?
- (22.6) *Gravitational clock effect:* Consider the Schwarzschild geometry describing the gravitational field of the Earth, which is rotating on its axis with angular velocity ω .
- (a) For an observer at rest on the Earth's surface at the equator, show that the proper time interval is given approximately by
- $$d\tau_1 \approx \left(1 - \frac{M}{r} - \frac{r^2\omega^2}{2}\right) dt. \quad (22.42)$$
- (b) Now consider a second observer who flies eastwards around the equator with velocity v at a height h , whose measures a proper time interval $d\tau_2$. Show that
- $$d\tau_1 - d\tau_2 \approx \left[-\frac{Mh}{r^2} + \frac{(2r\omega + v)v}{2}\right] dt. \quad (22.43)$$
- (c) Define $\Delta = (d\tau_1 - d\tau_2)/d\tau_1$ and estimate this quantity for an eastward flight around the Earth.

(d) Contrast the value of Δ for a similar flight in a westward direction.

This experiment was carried out in the early 1970s by Hafele and Keating, whose results were consistent with predictions in this problem [which follows the method in Ryder (2009)].

- (22.7) Two observers start on the same radial line in the Schwarzschild geometry, one at rest at r_1 and an-

other at rest at $r_2 > r_1$. At $t = 0$, the observer at r_2 begins to fall freely. What is the relative velocity of the observers as they meet?

Hint: Recall from Chapter 2 that γ , corresponding to the instantaneous relative velocity v_{rel} , can be found by taking the dot product of the two observers' velocity vectors: $-\gamma(v_{\text{rel}}) = \mathbf{u} \cdot \mathbf{v}$.